

Ahmed Ikram | CS Student | Full Stack & Data

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PROFILE

3rd-year Computer Science (Data Science & AI) student at the University of Dundee, currently on track for a First Class degree, seeking a software engineering or technology placement. Experienced in leading Agile teams and building full-stack systems across backend (FastAPI, C++), mobile (React Native), and data pipelines (ETL, ML, Power BI). Adept at translating complex technical requirements into deliverable solutions. Quick learner with active open-source contributions to kana-dojo and protein-drink-tracker.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027 (with year-long placement, June 2028)

PROJECTS

ATM Log Aggregation, Analysis, Diagnostics Platform | Python, FastAPI, SQLite, APScheduler, React, Vite

- Served as Team Lead and Scrum Master on a 7-person Agile team delivering an industry project for NCR Atleos; facilitated sprints and standups while owning backend, data engineering, and API delivery end-to-end
- Built a FastAPI system across 6 routers with JWT authentication and RBAC; designed a normalised SQLite data lake ingesting and validating logs from 7 ATM sources with structured error logging
- Delivered 48 automated tests across 8 subsections covering API contracts, pipeline correctness, and auth flows
- Currently extending with a RAG-based natural language diagnostic interface (LangChain, ChromaDB) and an event-driven ingestion pipeline with real-time Prometheus/Grafana observability

StockLens Mobile App | React Native Expo, TypeScript, Firebase, Node.js, Jest, React Native Testing Library

- Built a full-stack mobile app with Firebase authentication, device biometric locking, real-time sync, and an OCR receipt scanning pipeline that maps spending to stock tickers via the Alpha Vantage API - surfacing 20 years of historical returns and forward-looking CAGR projections
- Architected a modular TypeScript codebase with strict separation of concerns and a custom encryption layer for sensitive data at rest
- Wrote 78 Jest tests covering OCR parsing, CAGR math, crypto roundtrip, auth/lock flows, and screen-level user journeys
- Currently migrating to a FastAPI/PostgreSQL backend and extending with ML-driven spending analytics and deeper financial data integrations

W3C Web Logs ETL Pipeline | Apache Airflow, Python, Power BI, Power Automate, AWS RDS

- Designed an Apache Airflow DAG to extract, clean, and load raw web server logs into a Star Schema in PostgreSQL; handled IP-to-geolocation mapping, null handling, and field recoding in the transformation stage
- Built Power BI reports surfacing traffic trends, geographic distribution, and page-level engagement; presented findings as a recorded walkthrough video as well as a Power Automate automation which runs the entire ETL

AI Movie Recommendation System | Python, Scikit-learn, Flask

- Built a hybrid recommendation engine (collaborative filtering + content-based) on the MovieLens dataset, achieving up to ~78% hit rate and ~0.22 Precision@10
- Deployed via Flask with an interactive frontend; designed for extensibility with recommendation strategy injected as a dependency

Haggis Data Mining Project | Python, Scikit-learn, XGBoost, Jupyter Notebook

- Ran a complete ML pipeline from EDA through feature engineering to model evaluation; trained and compared 7 classifiers achieving ~90% classification accuracy, with clustering (K-Means, DBSCAN) used to validate natural groupings
- Translated findings into plain business recommendations supported by F1-scores, cross-validation, and residual diagnostics

PROJECTS

Rental Car Management System | C++

- Designed an OOP system with a polymorphic vehicle hierarchy, generic repository template pattern, and smart pointer memory management
- Implemented Levenshtein distance fuzzy search, automated late fee and loyalty rewards logic, and an 8-scenario built-in test suite with file-based persistence

TECHNICAL SKILLS

Languages	Python, Java, C++, C, JavaScript, TypeScript, SQL, Bash
Backend / APIs	Node.js, Flask, REST APIs, FastAPI, PostgreSQL, MySQL, MongoDB
Data / ML	Pandas, NumPy, Scikit-learn, PyTorch, Power BI, Jupyter, Apache Airflow
Cloud & DevOps	AWS (EC2, S3, RDS, SageMaker, IAM), Azure, Docker, Kubernetes, CI/CD, Linux (Ubuntu), Git
Web / Mobile	React Native Expo, HTML, CSS, Tailwind CSS, Bootstrap CSS, Firebase

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations	Feb 2025
AWS Academy Graduate - Cloud Foundations	Mar 2024
Foundations of Azure AI: Concepts, Capabilities, and Implementation	Oct 2024

EXPERIENCE

Beaulys Mini Market, Dundee, Scotland	
Retail Assistant	September 2023 - Present
- Troubleshooted and resolved occasional Point-of-Sale and stock system errors quickly to minimise customer wait time - Resolved customer inquiries and complaints in a fast-paced environment, maintaining a calm and professional demeanour	
Zulus Peri Peri Grill, Dundee, Scotland	
Front of House Worker	June 2021 - September 2023
- Trained 5 new staff members, helping them get up to speed quickly during busy service periods - Managed busy service periods by effectively prioritising orders, dine-in customers, and delivery orders, demonstrating strong time management under pressure	

TRANSFERRABLE SKILLS

Adaptability & Fast Learning Picked up React, FastAPI, and different python libraries independently across different projects.	Technical Communication Present technical work clearly to both technical peers and non-technical stakeholders.
Problem Decomposition Break complex technical challenges into manageable tasks with clear acceptance criteria.	Attention to Detail Thorough about testing, validation, and accuracy - in code, data, and documentation.
Teamwork & Collaboration Contributed to team projects and trained colleagues; comfortable in cross-functional settings.	Time Management Balanced university deadlines, part-time work, and independent projects simultaneously.
Initiative & Self-Motivation Built multiple personal projects outside coursework, including AWS and Azure certifications.	Growth Mindset Actively seek feedback, reflect on mistakes, and apply lessons to future work.